

Chapter - 4

4.1

4 bits input are 2 output are

(1) If 1 even are

(2) If 0 even output 1

(3) odd number of input even output 1

(a)

a	b	c	d	Q	
0	0	0	0	1	$\rightarrow a'b'c'd'$
0	0	0	1	1	$\rightarrow a'b'c'd$
0	0	1	0	1	$\rightarrow a'b'cd'$
0	0	1	1	0	
0	1	0	0	1	$\rightarrow a'b'cd$
0	1	0	1	0	
0	1	1	0	0	
0	1	1	1	1	$\rightarrow a'bcd$
1	0	0	0	1	$\rightarrow ab'c'd'$
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	1	$\rightarrow ab'cd$
1	1	0	0	0	
1	1	0	1	1	$\rightarrow abc'd$
1	1	1	0	1	$\rightarrow abcd'$
1	1	1	1	1	$\rightarrow abcd$

⑥ $F \Rightarrow a'b'c'd' + a'b'c'd + a'b'cd' + a'bc'd' + a'bcd + ab'c'd' + ab'cd + abc'd + abc d' + abcd$

		c		
		00	01	11 10
a	00	1	1	1
	01	1		1
	11		1	1
	10	1	1	1
		d		

} b

4.2 - output এর square value. 3 bit এর

7 অক্ষর 111, $7^2 \Rightarrow 49$ যা 110001, অক্ষর
6 bit value প্রদান।

	x	y	z	a	b	c	d	e	f	
0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	1	1
2	0	1	0	0	0	0	<u>1</u>	0	0	4
3	0	1	1	0	0	<u>1</u>	0	0	1	9
4	1	0	0	0	<u>1</u>	0	0	0	0	16
5	1	0	1	0	<u>1</u>	<u>1</u>	0	0	1	25
6	1	1	0	<u>1</u>	0	0	<u>1</u>	0	0	36
7	1	1	1	<u>1</u>	<u>1</u>	0	0	0	1	49

এই মান
গুলিতে
রেজাল্ট
বসবে

অক্ষর ~~z~~ এর f মান
e এর value 0

~~z~~ = f
e = 0

			1	1

z

$a \Rightarrow xy$ এর kmap

			1	
		1		

z

1	1	1		

z

$b \Rightarrow \cancel{xx} + \cancel{xx}y$ $xy' + x$

			1	
			1	

যে kmap $\Rightarrow yz'$

এর kmap $\Rightarrow xy' + x'yz \Rightarrow z(x \oplus y)$

$$a = xy$$

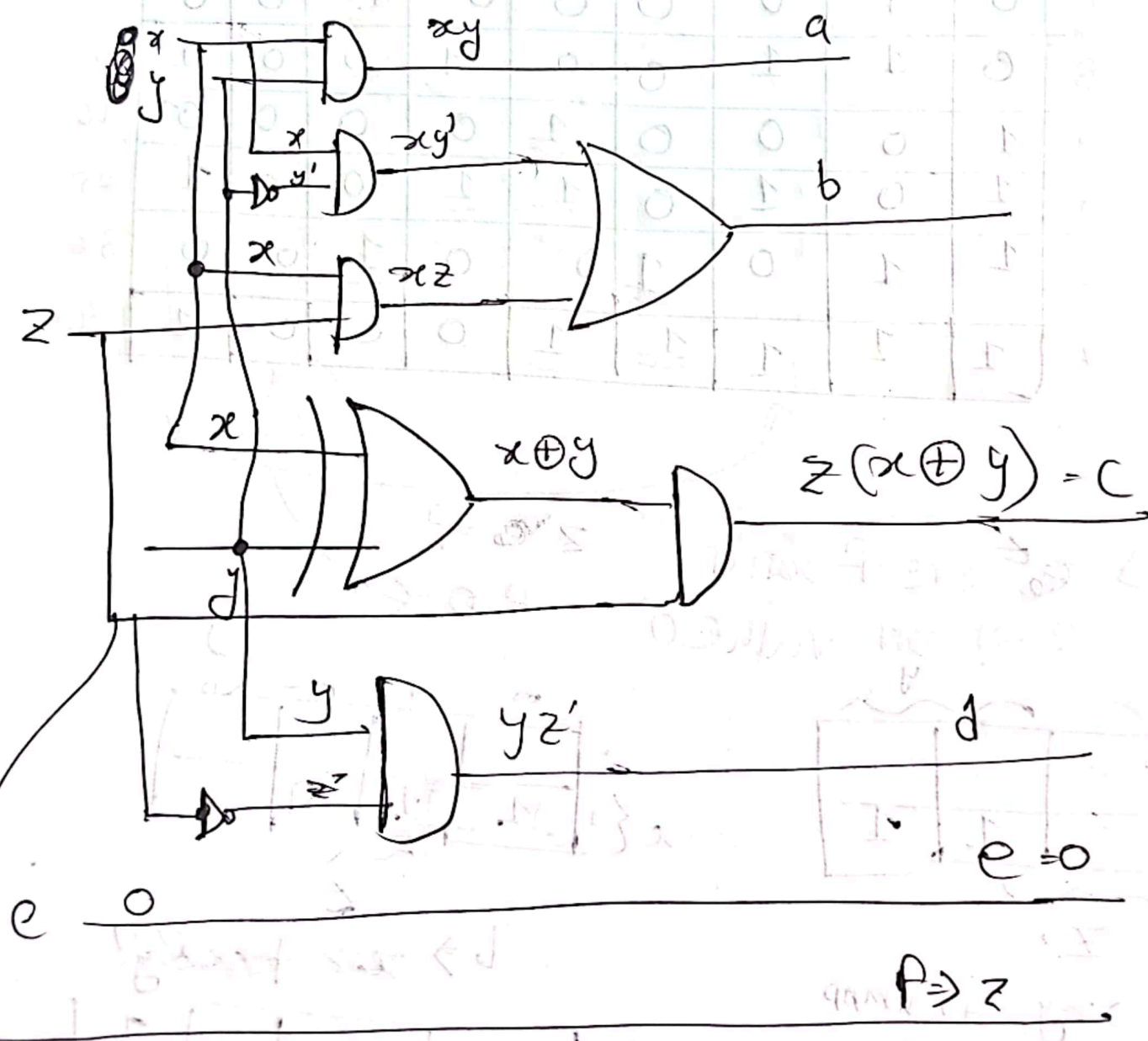
$$b \Rightarrow xy' + xz$$

$$c \Rightarrow z(x \oplus y)$$

$$d \Rightarrow yz'$$

$$e \Rightarrow 0$$

$$f \Rightarrow z$$



Q.3 2bit binary value to product

$A_1 \ A_0$

$B_1 \ B_0$

$A_1 B_0 \ A_0 B_0$

$A_1 B_1 \ A_0 B_1 \ X$

$C_2 \ A_1 B_1 \ A_1 B_0 \ A_0 B_0$
 $+ \ C_1 \ A_0 B_1$

$P_3 \ P_2 \ P_1 \ P_0$

OR gate or 1 bit half adder 2 bit binary add
 करके करे ।

$A_1 B_0 + A_0 B_1$ करके करे carry
 create 2 bit $A_1 B_1$ करके करे
 add 2 to 1

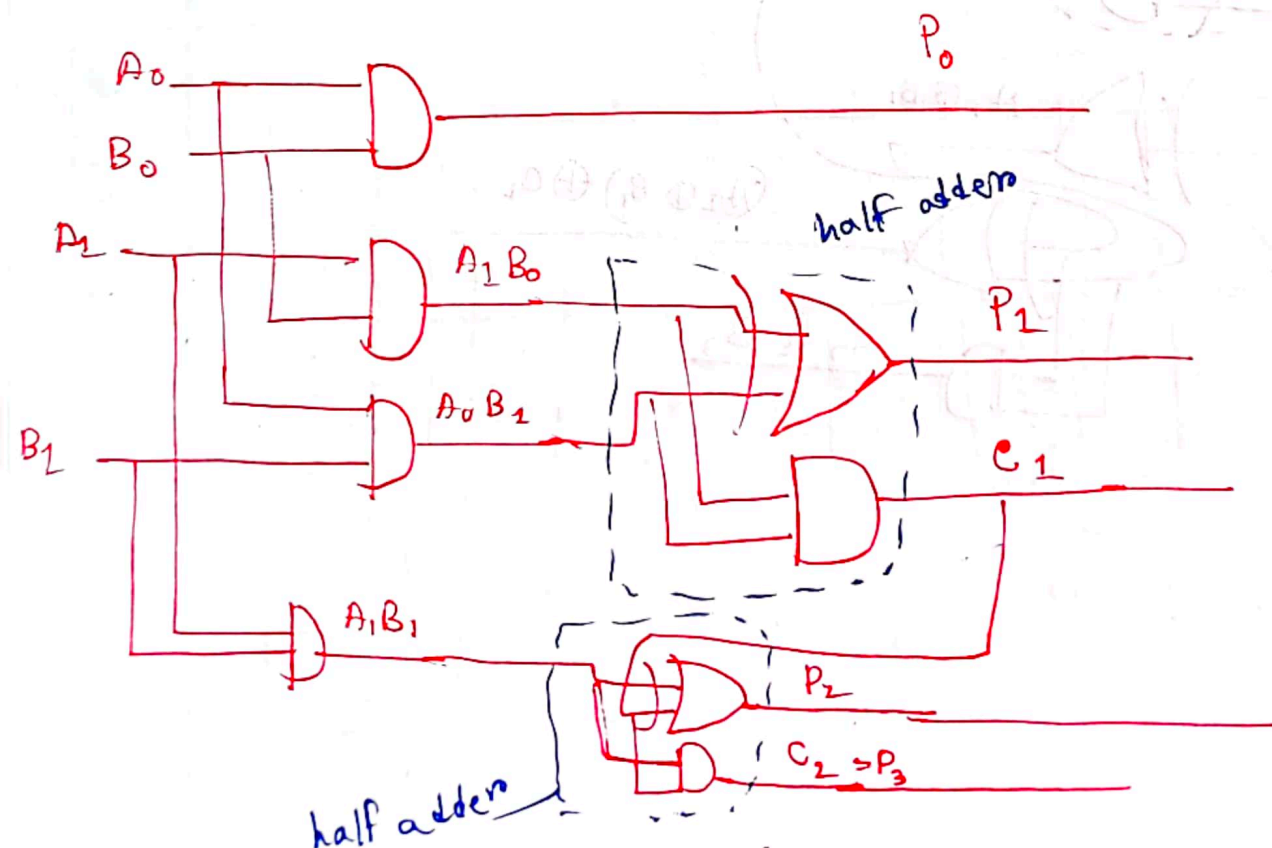
$$\therefore P_0 = A_0 B_0$$

$$P_1 \Rightarrow A_1 B_0 + A_0 B_1$$

$$P_2 \Rightarrow A_1 B_1 + C_1$$

$$C_2 \Rightarrow P_3$$

half adder



जबकि करी
 $P \Rightarrow x \oplus y$
 $C \Rightarrow xy$
 for half adder

2.4 2 bit binary sum $200 \Rightarrow$

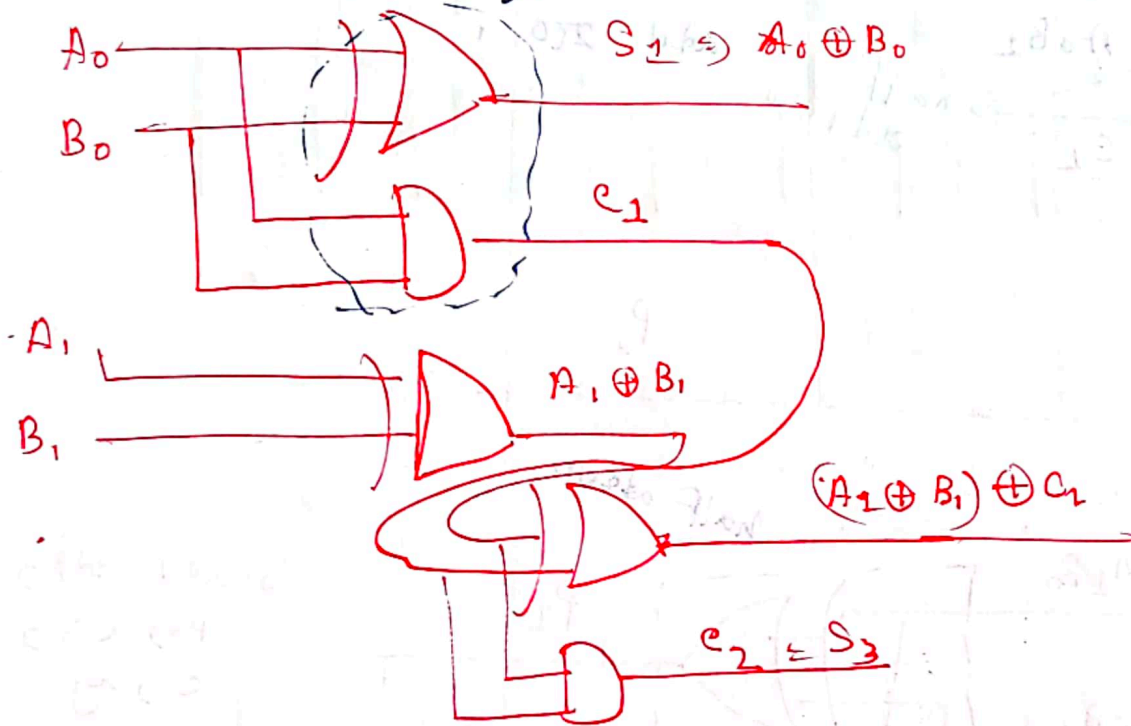
	A_2	A_1	A_0
c_2	B_2	B_1	B_0
$C_{2,1}$	$A_1 + B_1 + c_1$	$A_0 + B_0$	
$\downarrow$	$\downarrow$	$\downarrow$	
S_3	S_2	S_1	

$\therefore S_1 \Rightarrow A_0 + B_0$

$S_2 \Rightarrow A_1 + B_1 + \underline{1}$

$S_3 \Rightarrow C_3$

$\rightarrow$ half adders



4.5

Decimal number is 9's complement $\rightarrow$
 करके घटाना $\Rightarrow$ 9 minus करके घटाना (2011 $\Rightarrow$)

Decimal 7 is 9's complement 200

$$\begin{array}{r} 9 \\ -7 \\ \hline 2 \end{array} \rightarrow 9\text{'s complement}$$

BCD value to 9's complement $\rightarrow 100 \rightarrow 200 \rightarrow 9 \rightarrow 100$

BCD 27102 $\Rightarrow$ BCD 70 convert to minus 27102 to

$(2017) \rightarrow 9 \rightarrow 1001$

8 $\Rightarrow$ 1000

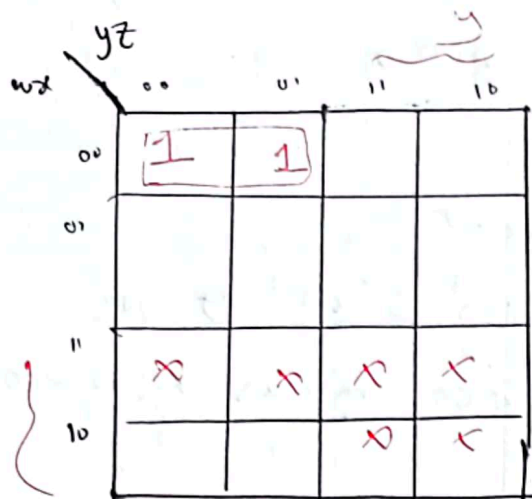
0001 $\rightarrow$ 9's complement

B C D

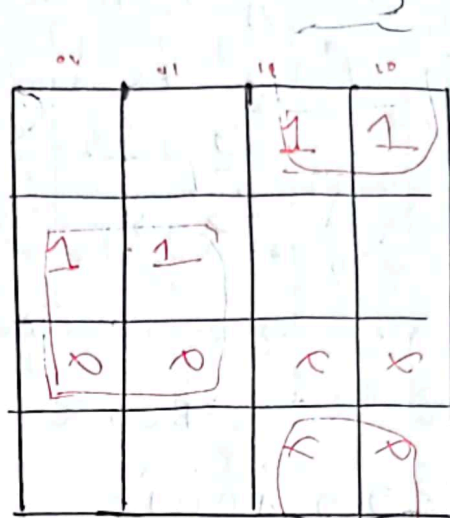
9's complement

	w	x	y	z	A	B	C	D	
0	0	0	0	0	1	0	0	1	$9-0 \Rightarrow 9$
1	0	0	0	1	1	0	0	0	$9-1 \Rightarrow 8$
2	0	0	1	0	0	1	1	1	$9-2 \Rightarrow 7$
3	0	0	1	1	0	1	1	0	$9-3 \Rightarrow 6$
4	0	1	0	0	0	1	0	1	$9-4 \Rightarrow 5$
5	0	1	0	1	0	1	0	0	$9-5=4$
6	0	1	1	0	0	0	1	1	$9-6 \Rightarrow 3$
7	0	1	1	1	0	0	1	0	$9-7=2$
8	1	0	0	0	0	0	0	1	$9-8 \Rightarrow 1$
9	1	0	0	1	0	0	0	0	$9-9=0$

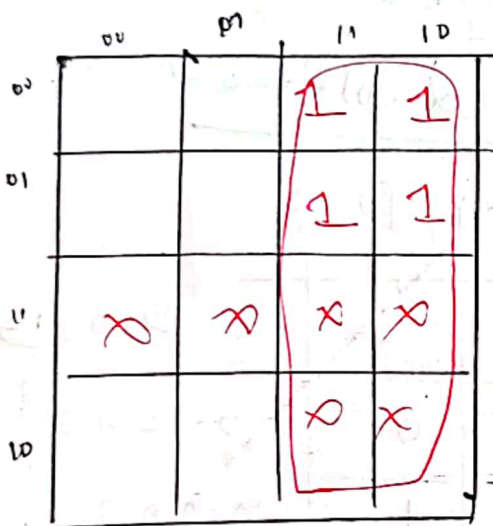
A, B, C, D of KMAP $\overline{x}y\overline{z}$ 500, 10-15 $\rightarrow$ don't care
 21100 1



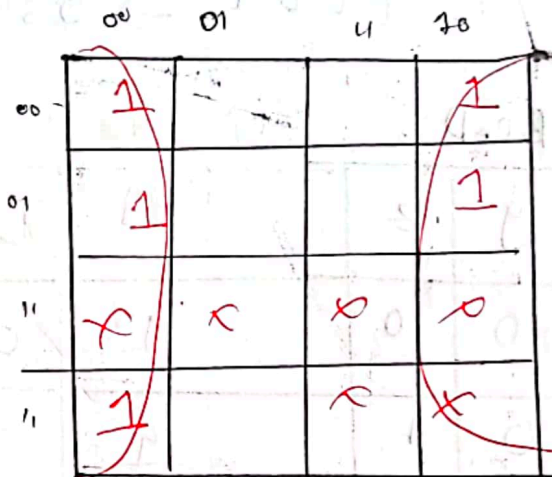
for $A \Rightarrow \overline{w}x'y'$



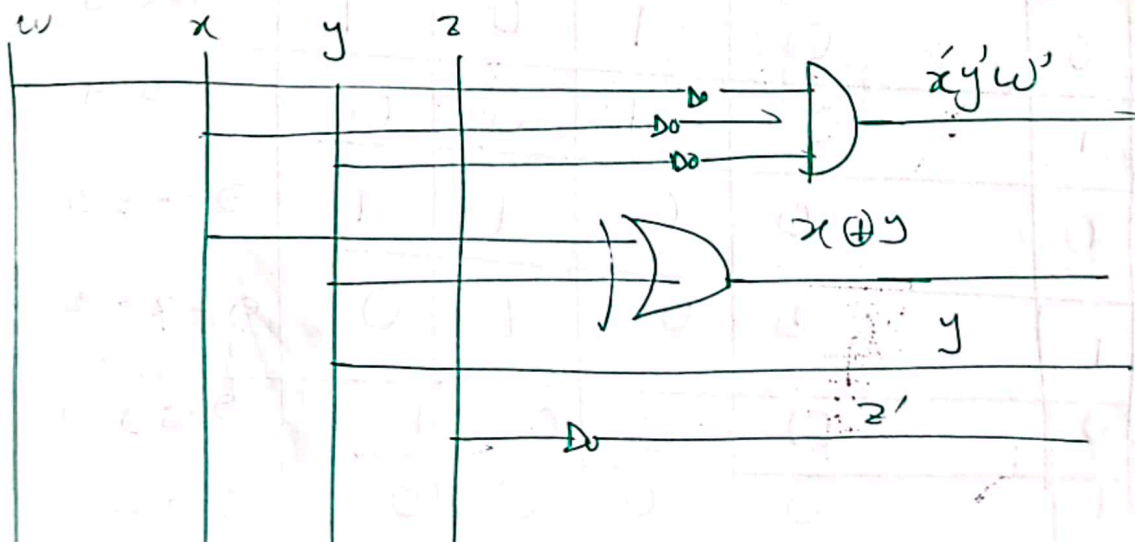
for $B \Rightarrow \overline{w}xy' + yz'x \Rightarrow x \oplus y$



for $C \Rightarrow y$



for $D \Rightarrow z'$



46

2's complement ~~10~~ 1's complement then

1 add $\Rightarrow$

Ans

~~1 1 1 1~~ 1's

1 $\rightarrow$ 0001

1's complement $\rightarrow$ 1 1 1 0 $\rightarrow$ ~~1 1 1 0~~ 2's complement

~~1 1 1 1~~

1 1 1 1

$\rightarrow$ 2's complement

	a	b	c	d	w	x	y	z
0	0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1
2	0	0	1	0	1	1	1	0
3	0	0	1	1	1	1	0	1
4	0	1	0	0	1	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	0
7	0	1	1	1	1	0	0	1
8	1	0	0	0	1	0	0	0
9	1	0	0	1	0	1	1	1
10	1	0	1	0	0	1	1	0
11	1	0	1	1	0	1	0	1
12	1	1	0	0	0	1	0	0
13	1	1	0	1	0	0	1	1
14	1	1	1	0	0	0	1	0
15	1	1	1	1	0	0	0	1

Binary value

2's complement

11 0 1
11 1 0

ab \ cd

	00	01	11	10
00		1	1	1
01	1	1	1	1
11				
10	1			

ab \ cd

	00	01	11	10
00		1	1	1
01	1			
11	1			
10		1	1	1

for $w \Rightarrow a'b + a'd + a'c + abc'd'$
 $\Rightarrow a'(b+d+c) + ab'c'd'$

$\Rightarrow a'(b+d+c) + a(b+c+d)'$

$w \Rightarrow a \oplus (b+d+c)$

$x \Rightarrow b'c + b'd + bc'd'$
 $\Rightarrow b'(c+d) + b(c+d)'$
 $x \Rightarrow b \oplus (c+d)$

ab \ cd

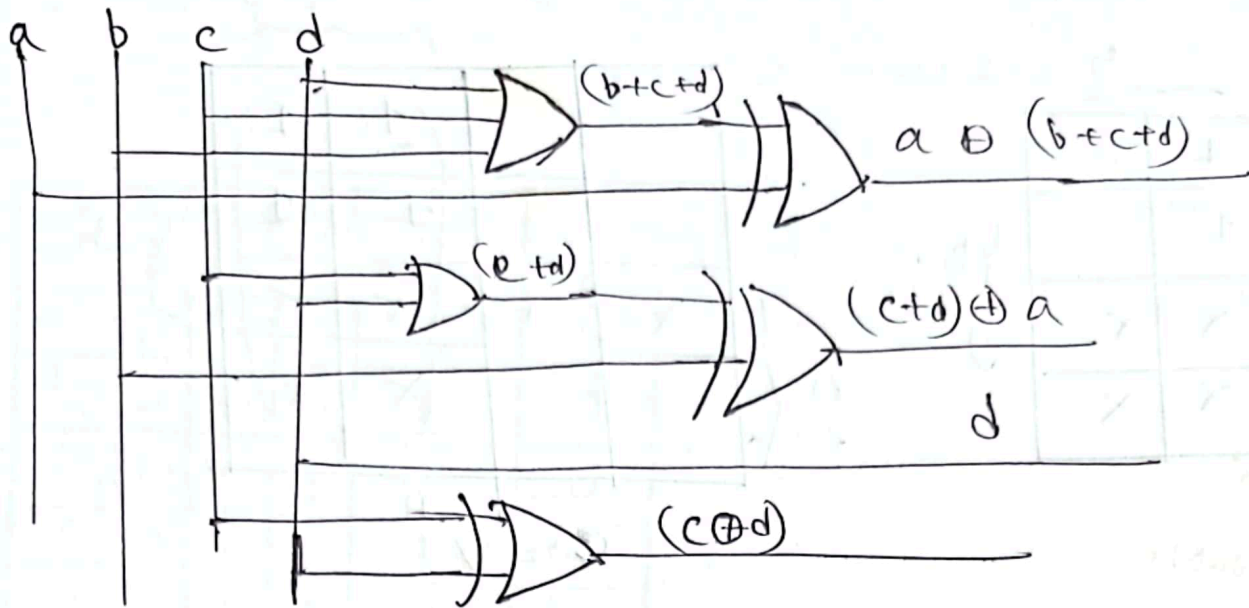
	00	01	11	10
00		1	1	
01		1	1	
11		1	1	
10		1	1	

$\Rightarrow d$

ab \ cd

	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

$\Rightarrow cd' + c'd \Rightarrow c \oplus d$



4.7

5 BCD হালত $\Rightarrow 0-9$ । So $9 \times 5 \Rightarrow 45$ যা represent করতে 8 bit লাগবে। অর্থাৎ 2 টি BCD $\Rightarrow$

Decimal (D)	BCD Input ($A_3 A_2 A_1 A_0$)	$5 \times D$	BCD Output (Tens digit, Units digit)
0	0000	00	0000 0000
1	0001	05	0000 0101
2	0010	10	0001 0000
3	0011	15	0001 0101
4	0100	20	0010 0000
5	0101	25	0010 0101
6	0110	30	0011 0000
7	0111	35	0011 0101
8	1000	40	0100 0000
9	1001	45	0100 0101

কারণ multiplies by 5 bolche

$5 \times 0 = 0$ er jnno 8 digit 0000 0000

$5 \times 1 = 05$ er jnno 8 digit 0000 0101

$5 \times 2 = 10$ er jnno 8 digit 0001 0000

input o BCD, Output o BCD
ar jehetu BCD tai 9 er porer gula Don't Care

AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11	x	x	x	x
10		1	x	x

AB \ CD	00	01	11	10
00			1	1
01			1	1
11	x	x	x	x
10			x	x

$D \Rightarrow$
from truth table

$$D = \bar{x} = \bar{z}$$

$$y = w = 0$$

AB \ CD	00	01	11	10
00	1	1	1	1
01	x	x	x	x
11			x	x

$$B = U$$

AB \ CD	00	01	11	10
00				
01				
11	x	x	x	x
10	1	1	x	x

$$A = T$$

If into input variable D is possible
line input variable D is possible

4.8

BCD এর তালিকা (০-৯) $\rightarrow$ valid $\rightarrow$ logic $\Rightarrow$ 0
 10-15 $\Rightarrow$ invalid $\Rightarrow$ logic $\Rightarrow$ 1

Decimal value	Input				Error Output (E)
	A	B	C	D	(Y)
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	0
10	1	0	1	0	1
11	1	0	1	1	1
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	1
15	1	1	1	1	1

যখন ইনপুট Valid (0000-1001), তখন Output = 0 (Logic 0)
 যখন ইনপুট Invalid (1010-1111), তখন Output = 1 (Logic 1)

valid, logic $\Rightarrow$ 0

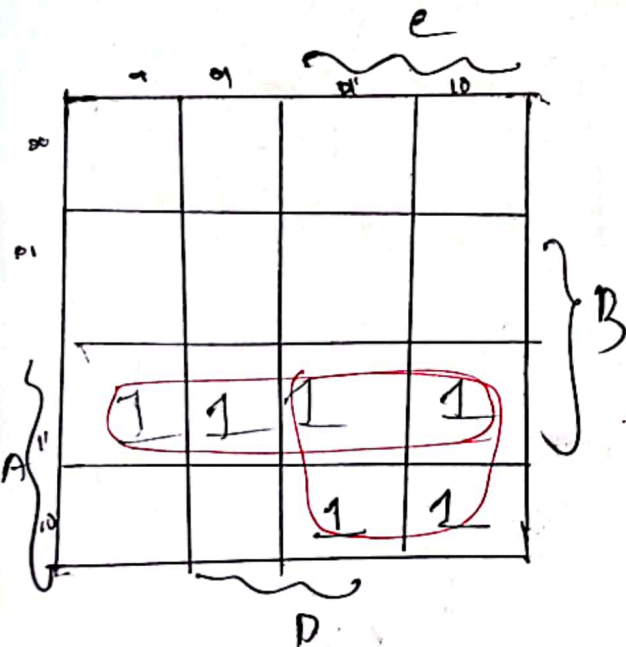
Unused Combinations (Invalid BCD Codes)
 চার বিটের সংখ্যা হলে মোট ১৬টি সম্ভাব্য মান (0000 থেকে 1111 পর্যন্ত) থাকতে পারে। কিন্তু BCD কোডে কেবলমাত্র ১০টি সংখ্যা ব্যবহৃত হয় (0000 থেকে 1001 পর্যন্ত)।

অবশিষ্ট ৬টি কম্বিনেশন (1010 - 1111) অবৈধ (Invalid BCD)।
 এগুলো থাকলে Error Detection Circuit একটিভ হবে (Logic 1)।

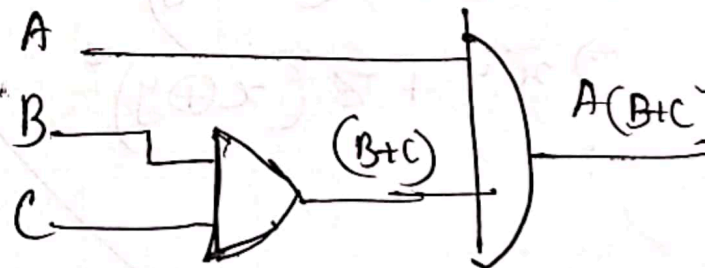
invalid - logic $\Rightarrow$ 1

সার্কিট ডিজাইন (Circuit Design)
 আমাদের লক্ষ্য একটি সার্কিট ডিজাইন করা, যা শুধুমাত্র অবৈধ BCD ইনপুটগুলোর জন্য ১ (Logic 1) আউটপুট দেবে।

KMAP করে ২টি বক্স



$\rightarrow AB + CA$
 $\Rightarrow A(B+C)$



D এর প্রয়োজন নেই

4.10.

full adder এর বৈজ্ঞানিক চিত্র

$$S \Rightarrow a \oplus b \oplus c$$

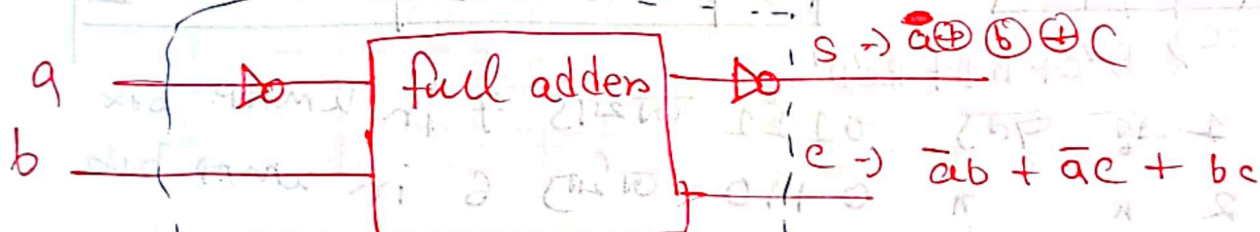
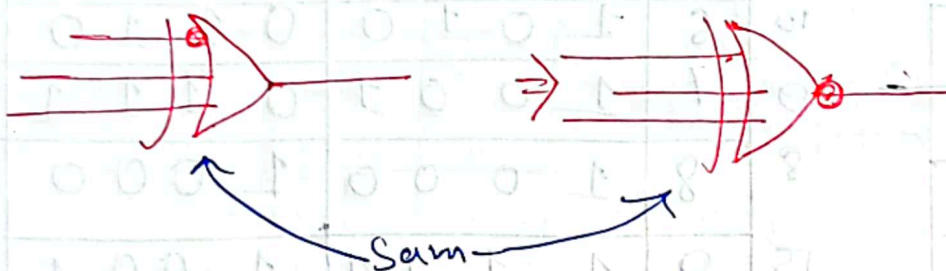
$$c \Rightarrow ab + bc + ac$$

Full subtractor এর বৈজ্ঞানিক চিত্র

$$D \Rightarrow a \oplus b \oplus c$$

$$b \Rightarrow \bar{a}b + \bar{a}c + bc$$

আমরা জানি,

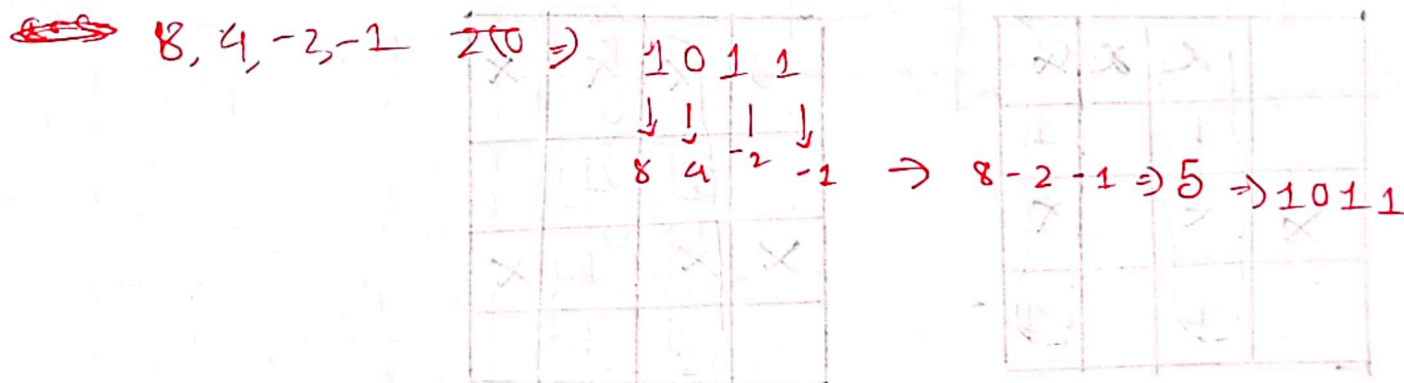


Full subtractors

4.11

8, 4, -2, -1 code converter প্রদান BCD তে convert

৪০৬ ২৮৪ ১ ৪ ৪, ৪, -২, -১ ৫ প্রমাণ ৫ যা



লাস্টে ভালো ভাবে করে দেয়া আছে

AB \ CD	00	01	11	10
00		X	X	X
01				
11	X	X	1	X
10	1			

for w $\Rightarrow 1 + B + A C' D'$

	X	X	X
1			
X	X		X
	1	1	1

for x $\Rightarrow B' C + B' D + B C' D'$

এখানে, 1 এর জন্য 0111 অর্থাৎ 7 in KMAP box
2 " " 0110 অর্থাৎ 6 in KMAP box

এখানে যে value আছে অর্থাৎ $\Rightarrow 0, 7, 6, 5, 4, 11, 10, 9, 8, 15$ মা valid.

যেগুলো নেই $\Rightarrow 1, 2, 3, 12, 13, 14$ ~~এইগুলো~~ মা invalid or don't care

Deci num	8	4	2	1	B C D
0	0	0	0	0	0 0 0 0
7	1	0	1	1	0 0 0 1
6	2	0	1	1	0 0 1 0
5	3	0	1	0	0 0 1 1
4	4	0	1	0	0 1 0 0
11	5	1	0	1	0 1 0 1
10	6	1	0	1	0 1 1 0
9	7	1	0	0	0 1 1 1
8	8	1	0	0	1 0 0 0
15	9	1	1	1	1 0 0 1

	X	X	X
	1		1
X	X		X
	1		1

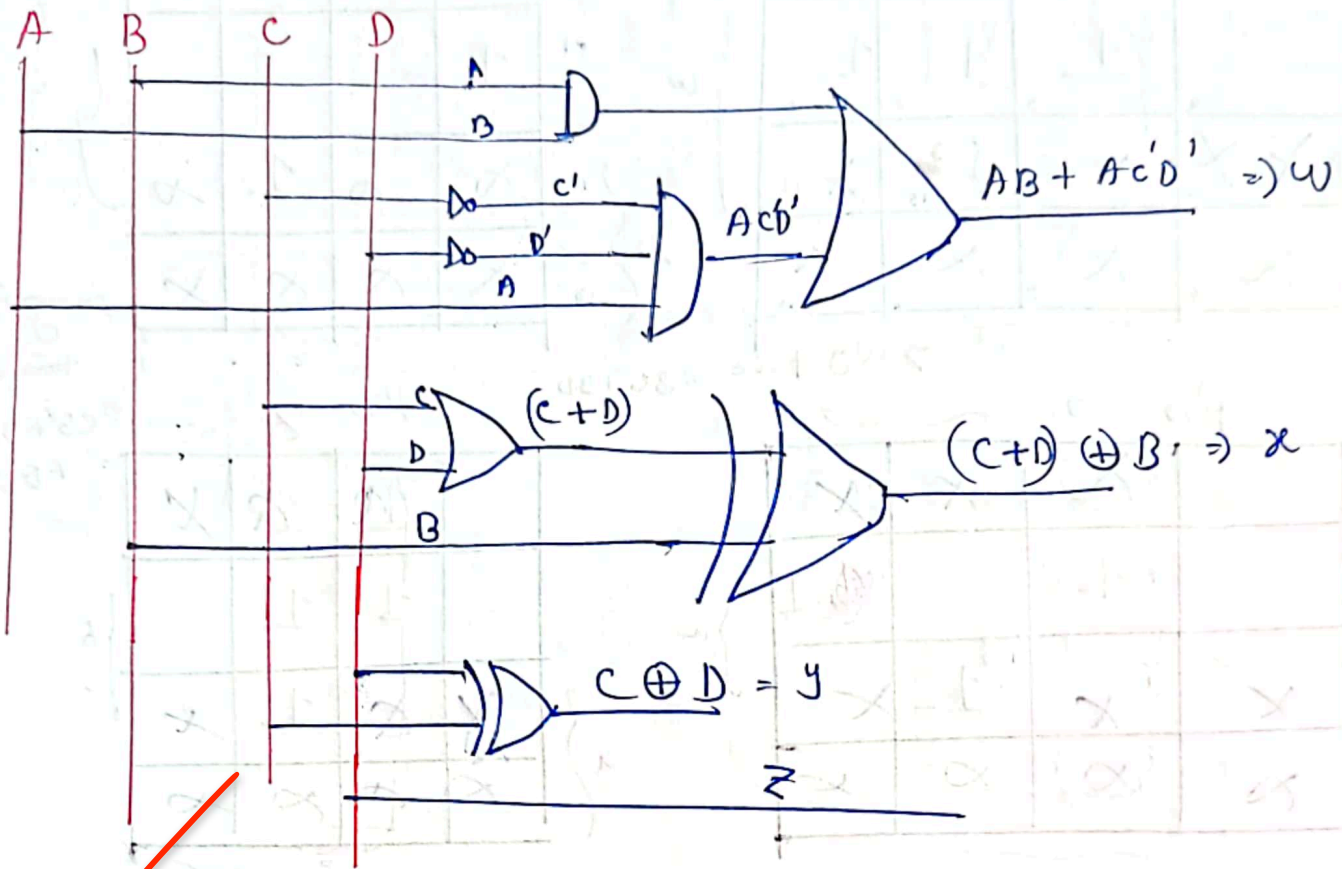
for y $\Rightarrow C' D + C D'$

	X	X	X
	1	1	
X	X	1	X
	1	1	

for z $\Rightarrow D$

$$W \Rightarrow AB + AC'D', \quad X \Rightarrow B'C + B'D + BC'D' \Rightarrow B'(C+D) + B(C+D)' \Rightarrow B \oplus (C+D)$$

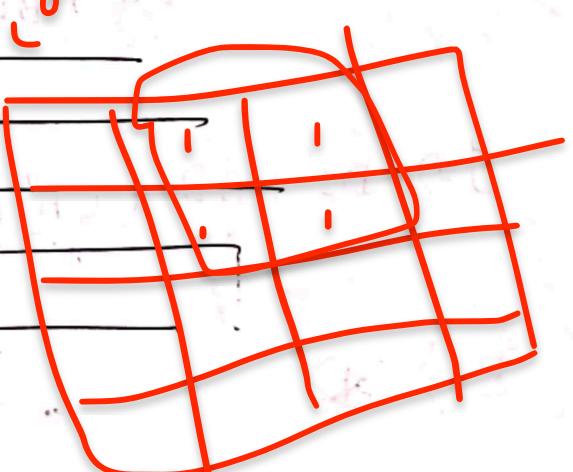
$$Y \Rightarrow C'D + CD' \Rightarrow C \oplus D$$

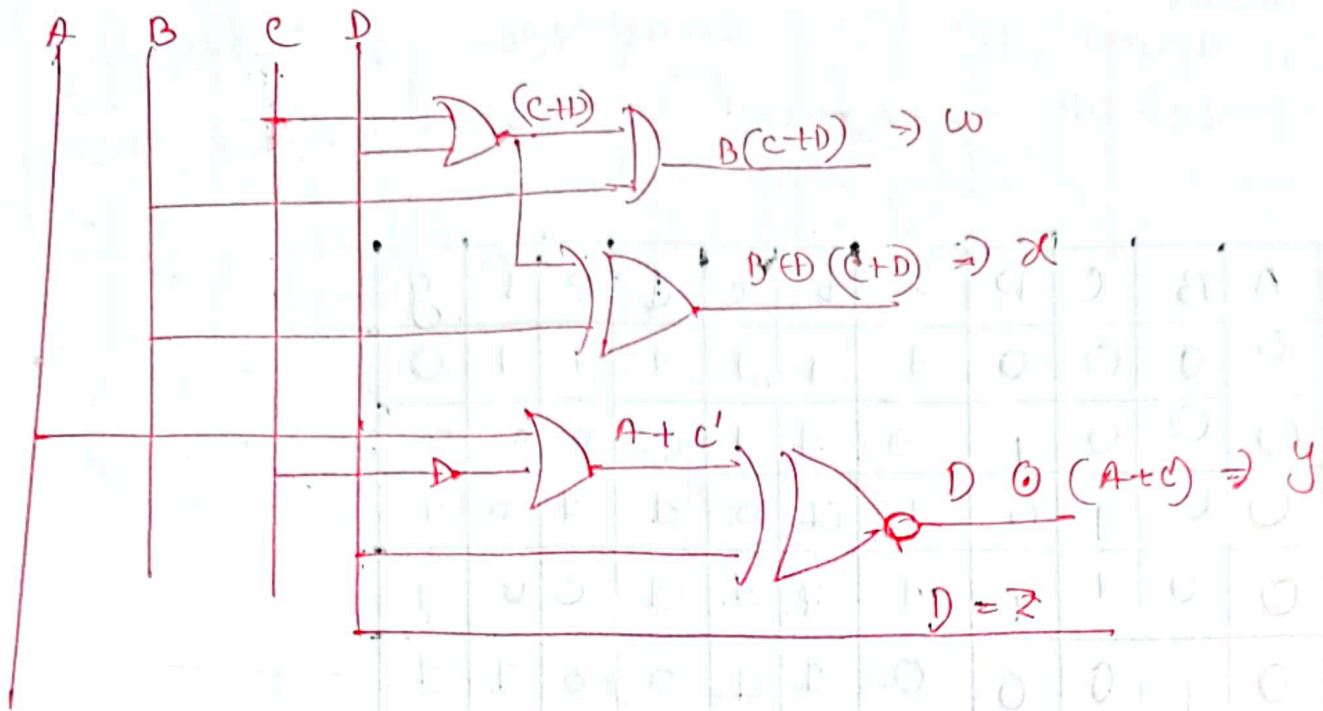


4.12

	Decimal	2	4	2	1	8	4	-2	-1
		a	b	c	d	w	x	y	z
0	0	0	0	0	0	0	0	0	0
1	1	0	0	0	1	0	1	1	1
2	2	0	0	1	0	0	1	1	0
3	3	0	0	1	1	0	1	0	1
4	4	0	1	0	0	0	1	0	0
5	5	0	1	0	1	1	0	1	1
6	6	0	1	1	0	1	0	1	0
7	7	0	1	1	1	1	0	0	1
14	8	1	1	1	0	1	0	0	0
15	9	1	1	1	1	1	1	1	1

$2(0)+4(0)+2(0)+1(1)=1$
 এখন 8 4 -2 -1 থেকে যোগ
 বিয়োগ করে 1 মিলাইতে হবে।
 $4-2-1=1$
 এর নিচে 1 বাকীগুলোতে শূন্য





1	0	0	1	0	0	0	0	1	0	1
1	0	1	1	0	1	1	0	1	0	2
1	1	1	1	0	1	0	1	1	0	3
0	0	0	0	1	1	1	1	1	0	4
1	1	1	1	1	1	0	0	0	1	5
1	1	0	1	1	1	1	0	0	1	6

1	1	1	1
	1		1
0	0	0	0
0	0	1	1

1	1		1
1	1	1	
0	0	0	0
0	0	1	1

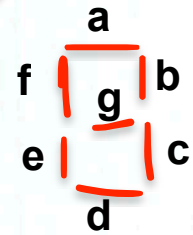
4.14

~~decimal~~

decimal to BCD convert করতে হবে এবং

BCD to 7 segment to convert করতে হবে

এখানে g মিসিং
মানে g=0
বাকীসব ১



যেটা মিসিং থাকবে সেটা ০ বাকীসব ১

Deci no	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	0	0	1	1	0	1
3	0	0	1	1	1	0	1	1	0	0	1
4	0	1	0	0	0	1	0	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

don't care to 0 ফাঁকি করতে led গুলো 10-15 line off
অকশ্যে আছে

1		1	1
	1	1	1
0	0	0	0
1	1	x	0

$$a \Rightarrow CA' + DA'B + AEB' + B'C'D'$$

1	1	1	1
1		1	
0	0	0	0
1	1	0	0

$$b \Rightarrow A'B' + C'PA' + CDA' + AC'B'$$

1	1	1	
1	1	1	1
0	0	0	0
1	1	0	0

$$C \Rightarrow BA' + C'A' + DA' + AB'C'$$

1		1	1
	1		1
0	0	0	0
1	1	0	0

$$d \Rightarrow B'A'C + CDA' + AB'C' + B'C'D' + DBA'C'$$

1			1
			1
0	0	0	0
1		0	0

$$e \Rightarrow A'ed' + B'c'd'$$

1			
1	1		1
0	0	0	0
1	1	0	0

$$f \Rightarrow AB'C' + BA'C' + BA'D' + A'c'd'$$

1			1
1	1		1
0	0	0	0
1	1	0	0

$$g \Rightarrow AB'C' + A'Bc' + A'B'C + A'cD'$$

$$a \Rightarrow A'(BD + C) + B'C'(A + D')$$

$$b \Rightarrow A'(CD + c'D') + B'(A' + Ac')$$

$$c \Rightarrow A'(B + C) + AB'C'$$

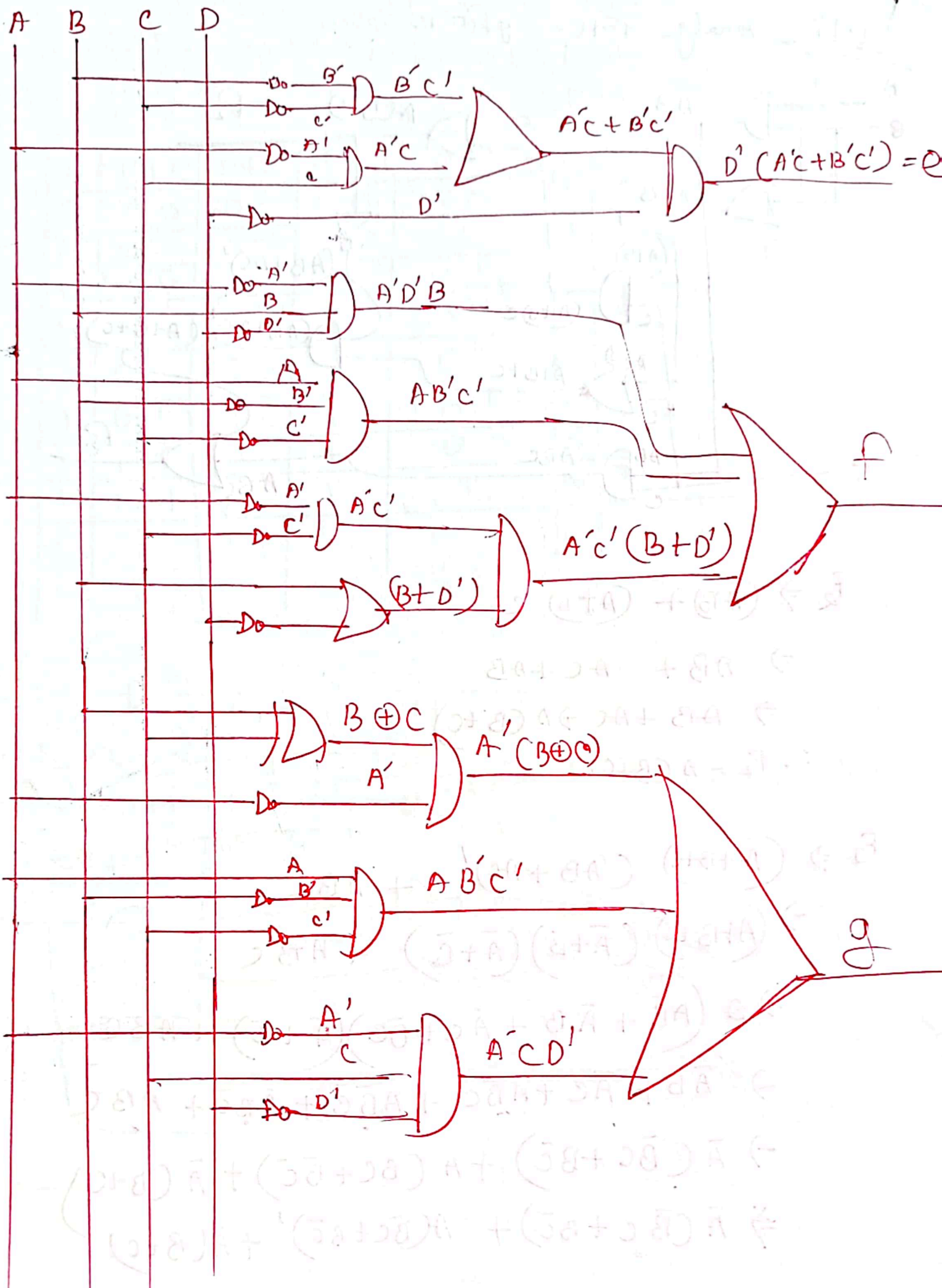
$$d \Rightarrow A'C(B' + D') + B'C'(A + D') + A'C'BD$$

$$e \Rightarrow D'(A'C + B'C')$$

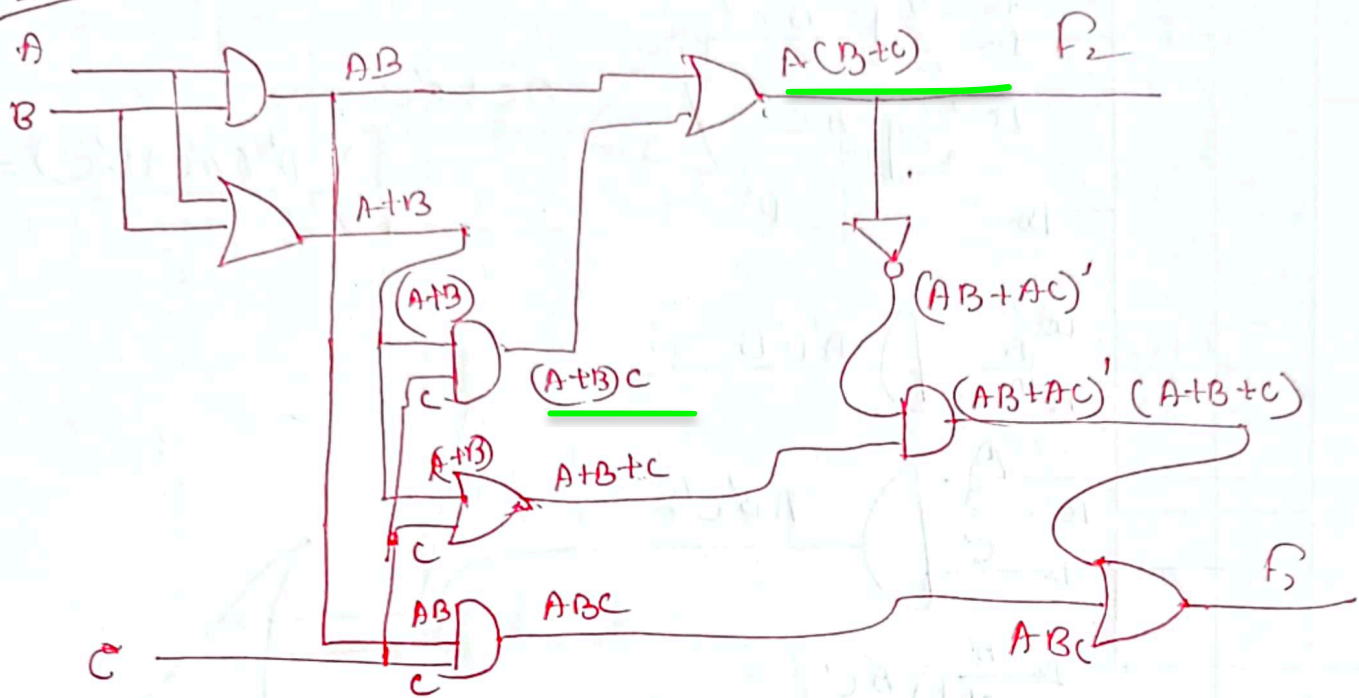
$$f \Rightarrow A'C'(B + D') + AB'C' + BA'D'$$

$$g \Rightarrow A'(B'C' + B'C) + AB'C' + A'cD'$$

$$\Rightarrow A'(B \oplus C) + AB'C' + A'cD'$$



4.15 Analyz द्वारे प्रतिपात ?



$$F_2 \Rightarrow (AB) + (A+B)C$$

$$\Rightarrow AB + AC + AB$$

$$\Rightarrow AB + AC \Rightarrow A(B+C)$$

$$\therefore F_2 = A(B+C)$$

$$F_3 \Rightarrow (A+B+C)(AB+AC)' + ABC$$

$$\Rightarrow (A+B+C)(\bar{A}+\bar{B})(\bar{A}+\bar{C}) + ABC$$

$$\Rightarrow (\bar{A}\bar{B} + \bar{A}B + \bar{A}C + \bar{B}C)(\bar{A}+\bar{C}) + ABC$$

$$\Rightarrow \bar{A}\bar{B} + \bar{A}C + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC$$

$$\Rightarrow \bar{A}(\bar{B}C + B\bar{C}) + A(\bar{B}C + B\bar{C})' + \bar{A}(B+C)$$

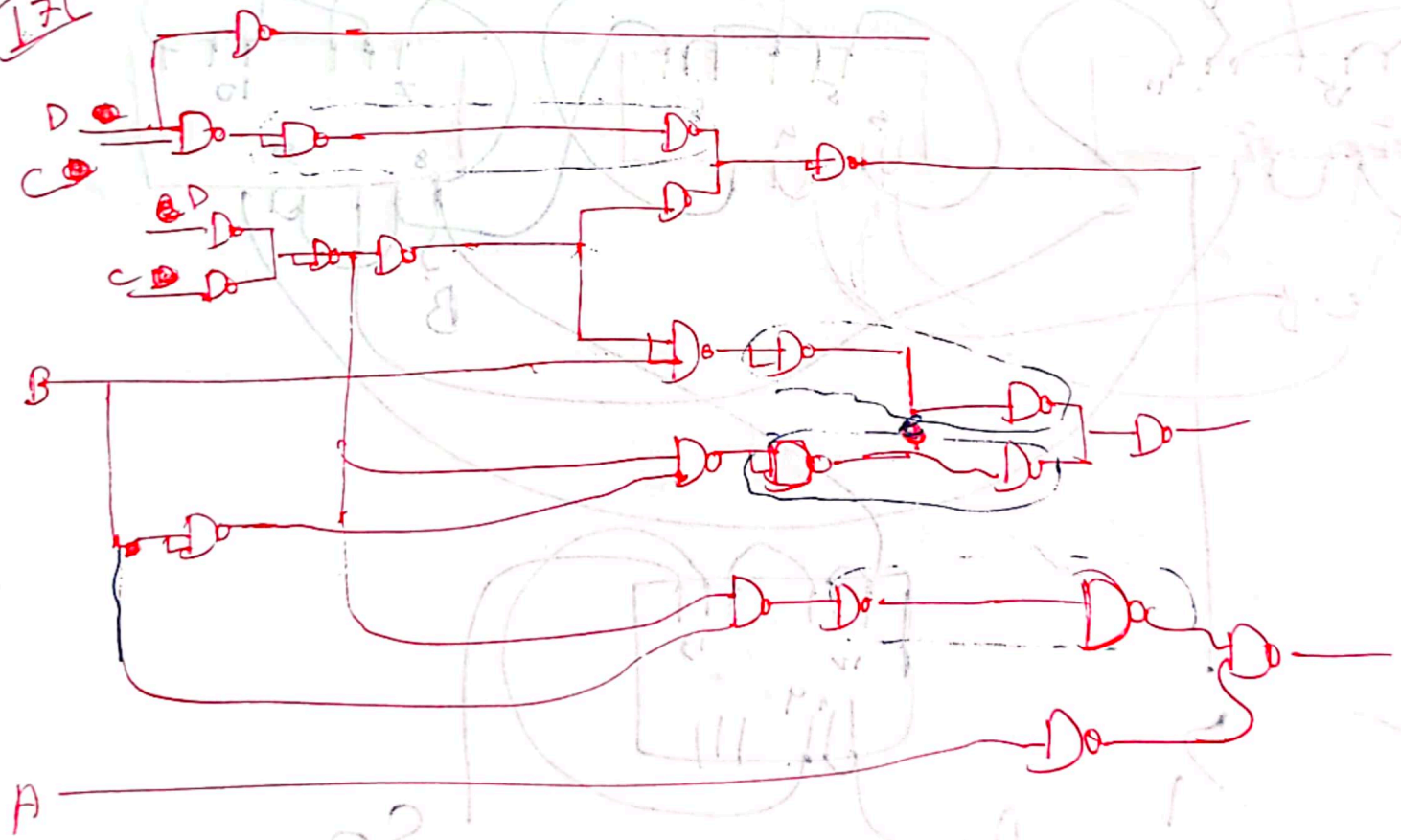
$$\Rightarrow \bar{A}(\bar{B}C + B\bar{C}) + A(\bar{B}C + B\bar{C})' + \bar{A}(B+C)$$

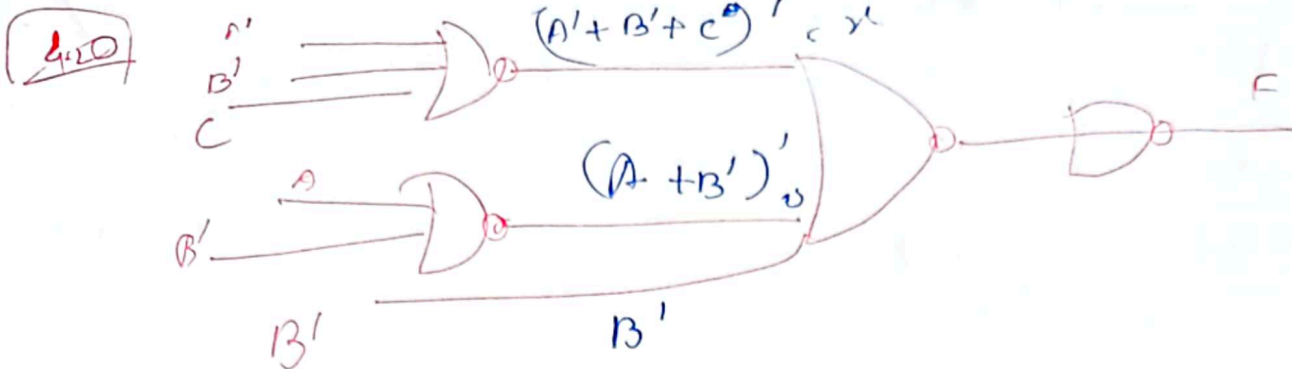
$$\rightarrow \underline{A \oplus B \oplus C} + \bar{A}(B+C) \Rightarrow F_1$$

4-16

F_2					$\neg F_1$			
A	B	C	$B+C$	$A(B+C)$	$A \oplus B \oplus C$	$\bar{A}$	$\bar{A}(B+C)$	$A \oplus B \oplus C + \bar{A}(B+C)$
0	0	0	0	0	0	1	0	0
0	0	1	1	0	1	1	1	1
0	1	0	1	0	1	1	1	1
0	1	1	1	0	0	1	1	1
1	0	0	0	0	0	0	0	0
1	0	1	1	1	0	0	0	0
1	1	0	1	1	0	0	0	0
1	1	1	1	1	1	0	0	1

17



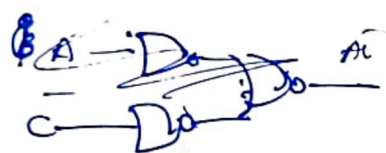


$$F \Rightarrow (A' + B' + C)' + (A + B')' + B'$$

$$\Rightarrow ABC' + \bar{A}B + \bar{B}$$

$$\Rightarrow B(AC + \bar{A}) + \bar{B}$$

$\Rightarrow$



$$x \Rightarrow ABC'$$

$$y \Rightarrow (A + B')$$

$$F = x + y$$

$$\Rightarrow (AC + B' + B')$$

$$\Rightarrow (AC + B')$$

$$\Rightarrow (A' + C)B$$

$$\Rightarrow B'B + B'C'$$

$$z \Rightarrow (x + y)$$

$$\Rightarrow ABC' + (A + B')$$

$$\Rightarrow ABC' + A'B$$

$$\Rightarrow B(AC' + A')$$

$$\Rightarrow B' + (A' + C)A$$

$$\Rightarrow B' + A + AC$$

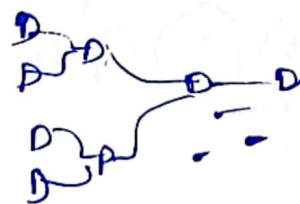
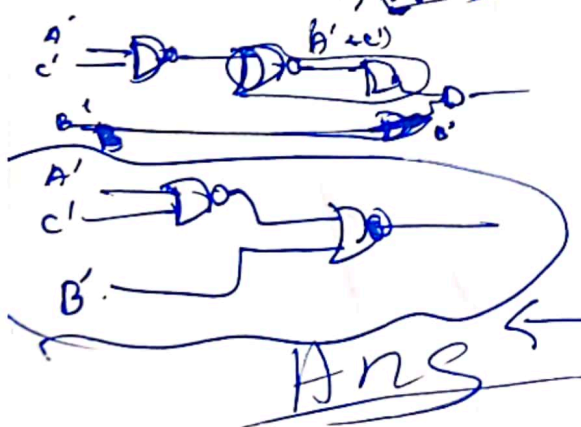
$$\Rightarrow AC + B'$$

$$\Rightarrow A'B$$

$$\therefore Z \Rightarrow AC + B'$$

$$F \Rightarrow \overline{Z + B'}$$

$$\Rightarrow (A' + C)B$$



9.19 C: $xy + yz + zx$

$S \Rightarrow C' (x+y+z) + xyz$

$\Rightarrow (x'y' + x'z' + y'z') (x+y+z) + xyz$
 $= x'(x+y) + z(x+y)'$

$\Rightarrow z'(xy + x'y) + z(x'y' + xy)$

$\Rightarrow x \oplus (x \oplus y)$

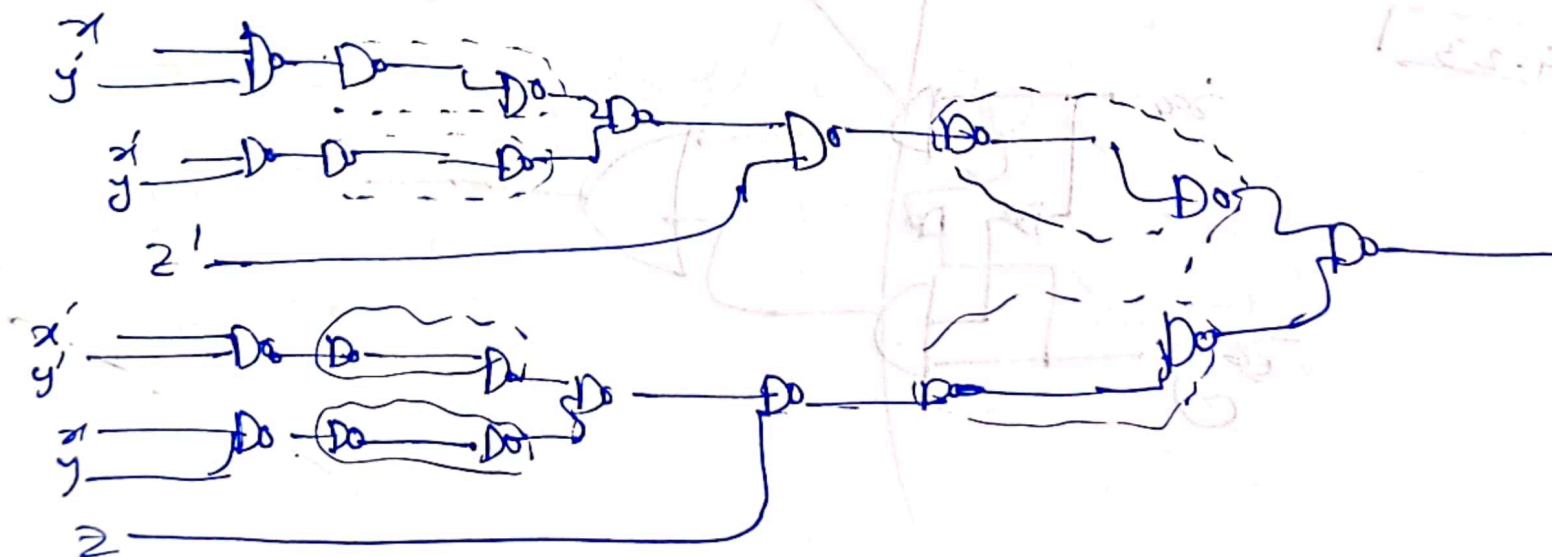
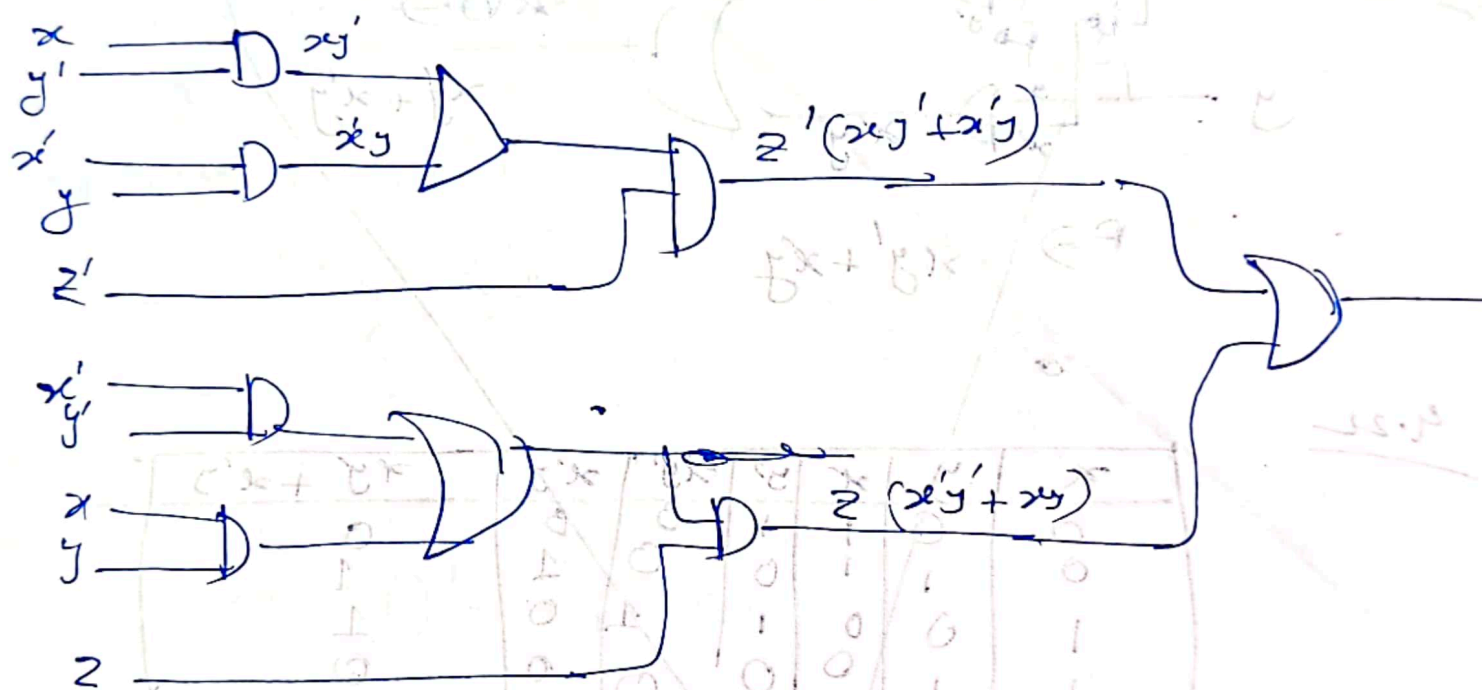
$C = (xy + yz + zx)'$

$\Rightarrow (x'+y')(x'+z')(y'+z')$

$\Rightarrow (x' + x'z' + y'x' + y'z') (y' + z')$

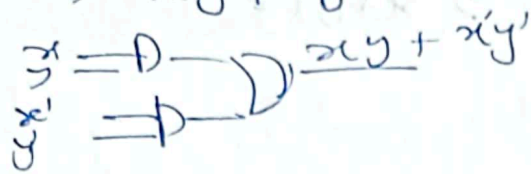
$\Rightarrow x'y' + x'z' + x'y'z' + x'z' + y'x' + x'y'z' + y'z' +$
 $x'z' + y'x' + x'y'z' + y'z' +$

$\Rightarrow x'y' + x'z' + y'z' = C$

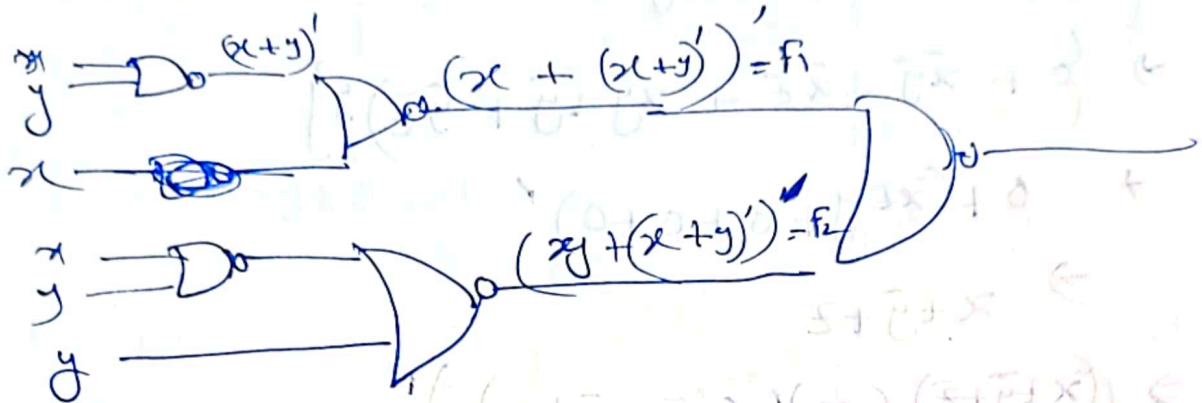


[25] two input equivalence $\Rightarrow$ ~~X~~ NOR

① XNOR $\Rightarrow xy + x'y'$



② XNOR $\Rightarrow$ NOR $\hookrightarrow$ convert



$$F_1 \Rightarrow (x + (x+y)')'$$

$$\Rightarrow x \cdot (x+y)$$

$$\Rightarrow x \cdot x + x \cdot y = x + xy$$

$$F_2 \Rightarrow (y + (x+y)')'$$

$$\Rightarrow y \cdot (x+y)$$

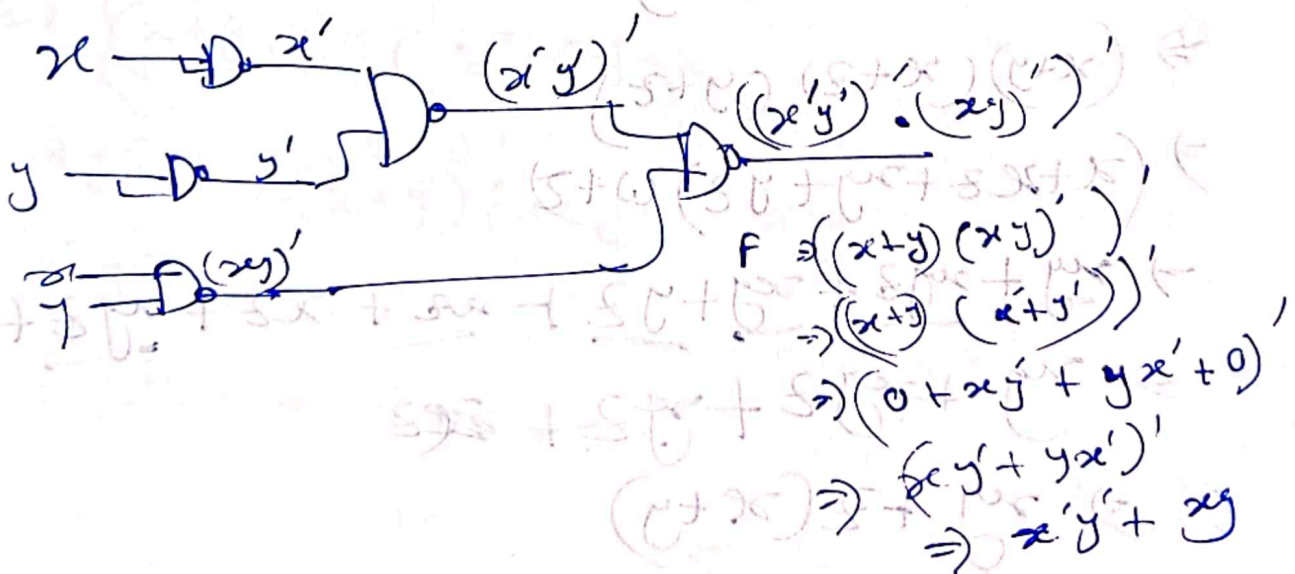
$$\Rightarrow xy + y \cdot y = xy + y$$

$$F = (F_1 + F_2)'$$

$$\Rightarrow (x + xy + xy + y)'$$

$$\Rightarrow (x + y)'$$

③ XNOR by NAND $\Rightarrow$



$$\underline{4.21} \quad F_1 \Rightarrow \cancel{x+y+z} \quad xz(xy')' \quad \Bigg| \quad F_2 = \cancel{yz}(xy')' \\ \Rightarrow x+y+z' \quad \Bigg| \quad \Rightarrow x+\bar{y}+\bar{z}$$

$$F_3 \Rightarrow \{ (xy)' \cdot (x+\bar{y}+\bar{z}) y \}'$$

$$\Rightarrow \{ (\bar{x}+y) (x+\bar{y}+\bar{z}) y \}'$$

$$\Rightarrow \{ 0 + \bar{x}\bar{y} + \bar{x}\bar{z} + x\bar{y} + y + y\bar{z} \}'$$

$$\Rightarrow 0 + (\bar{x}\bar{z}y + 0 + 0 + 0)'$$

$$\Rightarrow x+\bar{y}+\bar{z}$$

$$F_4 \Rightarrow ((x+\bar{y}+\bar{z})(xy)'(\bar{z}+\bar{x}+y))'$$

$$\Rightarrow \{ x\bar{z} + xy + \bar{y}\bar{z} + \bar{x}\bar{y} + \bar{z}\bar{x} + \bar{z}y \} (\bar{x}+\bar{y})'$$

$$\Rightarrow \cancel{0+0+0} \quad 0+0+(\bar{x}\bar{y}\bar{z} + \bar{x}\bar{y} + \bar{x}\bar{z} + x\bar{y}\bar{z} + x\bar{y}\bar{z} + 0 + \bar{y}\bar{z} + \bar{x}\bar{y} + \bar{x}\bar{y}\bar{z} + 0)'$$

$$\Rightarrow (\bar{x}\bar{y}\bar{z} + \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z} + \bar{x}\bar{y}\bar{z})'$$

$$\Rightarrow (\bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z})'$$

$$\Rightarrow (x+y)(x+z)(y+z)$$

$$\Rightarrow (x+yz+xz+xy+z)$$

$$\Rightarrow xy + xz + yz + x + xz + xy + yz$$

$$\Rightarrow xy + xz + yz + xz + xz$$

$$\Rightarrow xy + z(x+y)$$

$$F = \{x(x'y)(\bar{x} + \bar{z} + y)\}'$$

$$\Rightarrow \underline{x(x+y)}(\bar{x} + \bar{z} + y)'$$

$$\Rightarrow \underline{x\bar{y}}(\bar{x} + \bar{z} + y)$$

$$\Rightarrow 0 + (x\bar{y}\bar{z} + \cancel{0} + \cancel{xy})'$$

$$\Rightarrow \bar{x} + y + z$$

$$F = \{(x+y+z)(x+\bar{y}+\bar{z})z\}'$$

$$\Rightarrow \{(x\bar{z} + y\bar{z})(x+\bar{y}+\bar{z})\}'$$

$$= (\bar{x}\bar{z}\bar{y} + 0 + xy\bar{z})'$$

$$\Rightarrow (x+y+\bar{z})(\bar{x}+\bar{y}+\bar{z})$$

$$\Rightarrow 0 + x\bar{y} + x\bar{z} + \bar{x}y + \bar{z}y + \bar{x}\bar{z} + y\bar{z} + \bar{z}$$

$$\Rightarrow \bar{x}\bar{y}\bar{z} + \bar{z}$$

$$\Rightarrow \bar{z}' + \bar{z}' + \bar{z}' + \bar{x}\bar{y} + \bar{x}y$$

$$\Rightarrow \bar{z} + \bar{x}\bar{y} + \bar{x}y$$

$$F_2 = \{(x+y+z)(\bar{z} + x\bar{y} + x\bar{y})(\bar{x} + y + z)\}'$$

$$\Rightarrow \bar{x}y\bar{z} + \bar{z}(\bar{x} + y)(x + \bar{y}) + x\bar{y}\bar{z}$$

$$\Rightarrow \bar{x}y\bar{z} + \bar{z}(\bar{x} + y)(x + \bar{y}) + x\bar{y}\bar{z}$$

$$\Rightarrow \bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + \bar{z}y(x + \bar{x}) + \bar{z}y\bar{x}$$

$$F_1 = ? \text{ do it}$$

	0	1	0	1
x\y	1	1	0	1

Not possible

$$\bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z}$$

$$\Rightarrow \bar{x}y\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z} = F_1$$

For p4-21 (b)

$$\begin{aligned} x &\Rightarrow ((A+B)' + A)' \\ &\Rightarrow (\overline{A+B} + A)' \\ &\Rightarrow (A+B) \cdot \overline{A} \\ &\Rightarrow \overline{A} B \end{aligned}$$

$$\begin{aligned} &((A+B)' + A)' \\ &\Rightarrow (A+B) \cdot \overline{A} \\ &\Rightarrow \overline{A} B \end{aligned}$$

$$\begin{aligned} x &\Rightarrow (\overline{A} B + A \overline{B} + C)' \\ &\Rightarrow (A+B) (\overline{A} + \overline{B}) C' \\ &\Rightarrow 0 + (\overline{A} B + A \overline{B}) C' \\ &\Rightarrow \overline{A} B C' + A \overline{B} C' \end{aligned}$$

$$\begin{aligned} &(\overline{A} B + \overline{A} \overline{B} + A B \overline{C} + A \overline{B} C)' \\ &\Rightarrow (\overline{A} + B) (A + \overline{B}) (\overline{A} + \overline{B} + C) \\ &\quad (A + B + C) \end{aligned}$$

$$\begin{aligned} &\Rightarrow (\overline{A} \overline{B} + B A) (\overline{A} B + \overline{A} \overline{B} + A B \overline{C} + A \overline{B} C + C) \\ &\quad \overline{A} \overline{B} + C \end{aligned}$$

$$\begin{aligned} &(\overline{A} B C + \overline{A} \overline{B} C + C)' \\ &\Rightarrow (\overline{A} + \overline{B} + C) (A + B + \overline{C}) C' \\ &\Rightarrow (\overline{A} B + \overline{A} \overline{B} + A \overline{B} + \overline{B} \overline{C} + A C + \overline{A} B) C' \end{aligned}$$

$$\begin{aligned} &\Rightarrow \overline{C} B + \overline{A} \overline{C} + \overline{A} \overline{B} \overline{C} + \overline{B} \overline{C} + A C + \overline{A} B C' \\ &\quad 0 + \overline{A} B C' \end{aligned}$$

$$\Rightarrow \overline{C} + A \overline{C} + A \overline{C}$$

$$\Rightarrow \overline{C} + A \overline{C}$$

$$\Rightarrow (\overline{C})$$

$$\Rightarrow \overline{A} \overline{B} \overline{A} \overline{B} C +$$

$$\Rightarrow (\overline{A} \overline{B} + \overline{A} B) (\overline{A} B + \overline{A} \overline{B} + C)$$

$$\Rightarrow \overline{A} \overline{B} C + A B C$$

$$\Rightarrow$$

$$(\overline{A} B C + \overline{A} \overline{B} C + C)'$$

$$\Rightarrow (\overline{A} + \overline{B} + C) (A + B + \overline{C}) C$$

$$\Rightarrow (\overline{A} B + \overline{A} \overline{B} + \overline{A} \overline{B} A + \overline{B} \overline{C} + \overline{C} A + \overline{C} B) C$$

$$\Rightarrow (\overline{C} + \overline{A} B + A \overline{B} + \overline{B} \overline{C} + \overline{C} B) C$$

$$\Rightarrow 0 + \overline{A} B C + A \overline{B} C + 0 + 0$$

$$\Rightarrow C (\overline{A} B + A \overline{B}) = F$$

9.22 truth table for P4-21 (a), (b)
do it

9.23

D

26

		obtained		known	
x	y	$x \oplus y$		$x \oplus y$	
0	0	0		0	
0	1	1		1	
1	0	1		1	
1	1	0		0	

XOR is true same time
 false time 1

27

$$F \Rightarrow \underline{A \oplus B \oplus C \oplus D}$$

$$\Rightarrow (\underline{AB + \bar{A}\bar{B}}) \oplus (\underline{CD + \bar{C}\bar{D}})$$

$$\Rightarrow (\underline{AB + \bar{A}\bar{B}}) (\underline{CD + \bar{C}\bar{D}}) + (\underline{AB + \bar{A}\bar{B}})' (\underline{CD + \bar{C}\bar{D}})'$$

$$\Rightarrow ABCD + AB\bar{C}\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}\bar{D} + (\bar{A} + \bar{B})(A + B)(\bar{C} + \bar{D})(C + D) + (\bar{A}B + A\bar{B})(\bar{C}D + C\bar{D})$$

$$\Rightarrow \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D}$$

$\left. \begin{matrix} 00 \\ 01 \\ 11 \\ 10 \end{matrix} \right\} A$
 $\left. \begin{matrix} 00 & 01 & 10 & 11 \end{matrix} \right\} B$

00	0		1	
01		1		1
11	1		1	
10		1		1

D

$$\Rightarrow \Sigma(0, 3, 6, 9, 10, 12, 15)$$

~~28~~

x	y	z	w	A	B	C	D	
0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	1
0	0	1	1	0	0	1	0	2
0	0	1	0	0	0	1	1	3
0	1	0	0	0	1	0	0	4
0	1	0	1	0	1	0	1	5
0	1	1	1	0	1	1	0	6
0	1	1	0	0	1	1	1	7
1	1	0	0	1	0	0	0	8
1	1	0	1	1	0	0	1	9
1	1	1	1	1	0	1	0	10
1	1	1	0	1	0	1	1	11
1	0	0	0	1	1	0	0	12
1	0	0	1	1	1	0	1	13
1	0	1	1	1	1	1	0	14
1	0	1	0	1	1	1	1	15

28

A	B	C	D	x	y	z	w
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	0 1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

1	1	1	1
1	1	1	1

for $w \Rightarrow A$

1	1	1	1
1	1	1	1

for $x = A\bar{B} + \bar{A}B$
 $\Rightarrow A \oplus B$

		1	1
1	1		
		1	1
1	1		

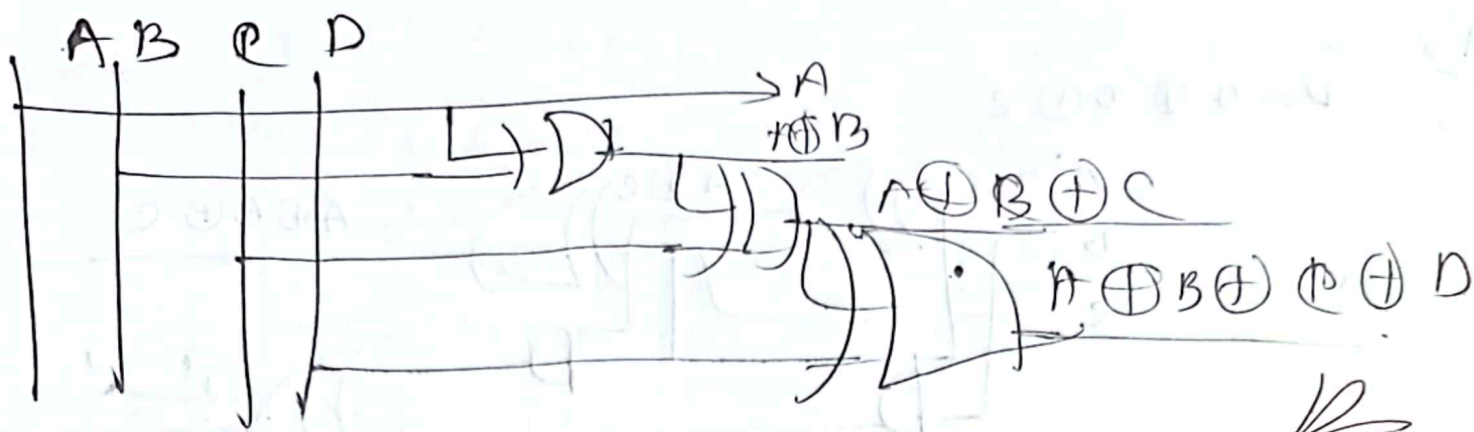
for $y \Rightarrow \bar{A}\bar{B}C + \bar{A}B\bar{C} +$
 $ABC + A\bar{B}\bar{C}$

$$\Rightarrow \bar{A}(B \oplus C) + A(B \oplus C)$$

$$\Rightarrow A \oplus B \oplus C$$

		1		1
1			1	
		1		
1			1	

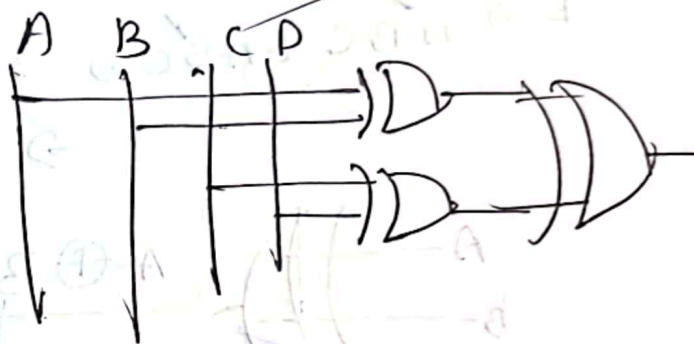
for $z \Rightarrow A \oplus B \oplus C$
 By simplifying



#. 29

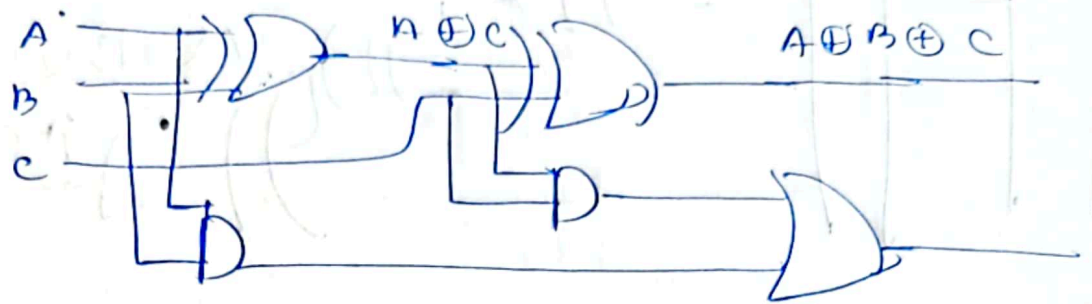
A	B	C	D	P
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

$$A \oplus B \oplus C \oplus D$$

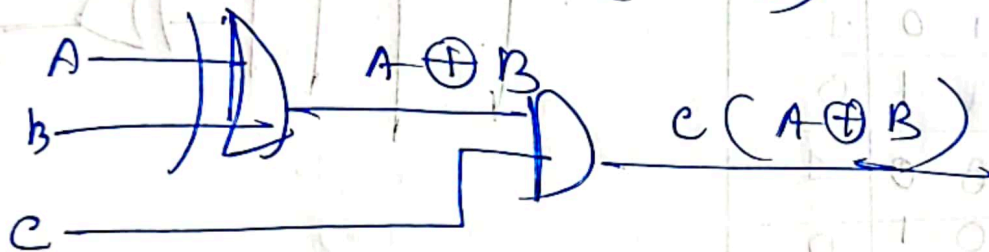


4.30

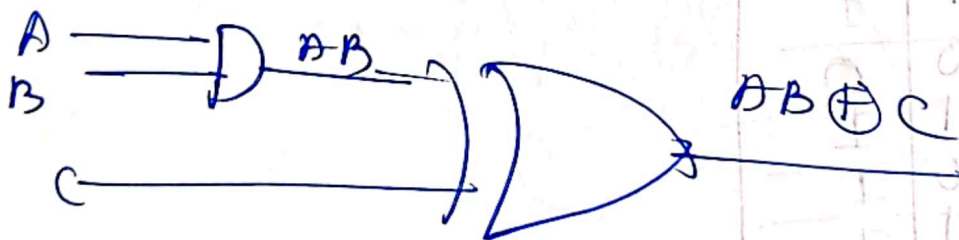
$$D = A \oplus B \oplus C$$



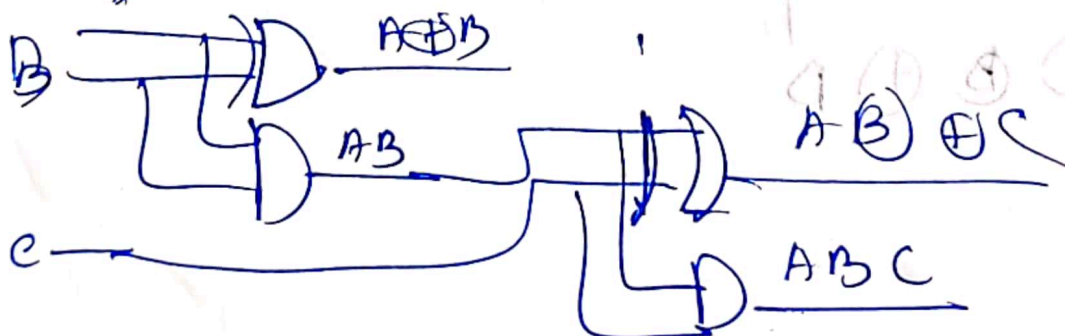
$$\begin{aligned} E &\Rightarrow A'BC + AB'C \Rightarrow c(A'B + AB') \\ &\Rightarrow c(A \oplus B) \end{aligned}$$



$$\begin{aligned} F &\Rightarrow ABC' + (A' + B')C \\ &\Rightarrow \cancel{AB}C' + (AB)'C \\ &\Rightarrow AB \oplus C \end{aligned}$$



$$G \Rightarrow ABC$$



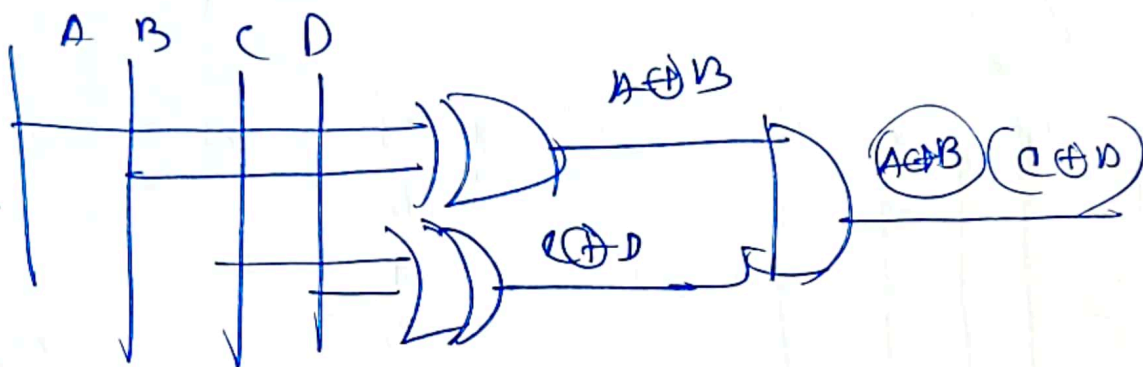
Q3)

$$F \Rightarrow AB'CD' + A'BCD' + AB'C'D + A'B'C'D$$

$$\Rightarrow CD'(AB' + A'B) + C'D(AB' + A'B)$$

$$\Rightarrow CD'(A \oplus B) + C'D(A \oplus B)$$

$$\Rightarrow (A \oplus B)(C \oplus D)$$



4.11

Step 2: Creating the Conversion Table

8-4-(-2)-(-1) Code (A B C D)	Decimal Value	BCD Output (W X Y Z)
0000	$8(0) + 4(0) - 2(0) - 1(0) = 0$	0000 (0)
0001	$8(0) + 4(0) - 2(0) - 1(1) = -1$	Invalid
0010	$8(0) + 4(0) - 2(1) - 1(0) = -2$	Invalid
0011	$8(0) + 4(0) - 2(1) - 1(1) = -3$	Invalid
0100	$8(0) + 4(1) - 2(0) - 1(0) = 4$	0100 (4)
0101	$8(0) + 4(1) - 2(0) - 1(1) = 3$	0011 (3)
0110	$8(0) + 4(1) - 2(1) - 1(0) = 2$	0010 (2)
0111	$8(0) + 4(1) - 2(1) - 1(1) = 1$	0001 (1)
1000	$8(1) + 4(0) - 2(0) - 1(0) = 8$	1000 (8)
1001	$8(1) + 4(0) - 2(0) - 1(1) = 7$	0111 (7)
1010	$8(1) + 4(0) - 2(1) - 1(0) = 6$	0110 (6)
1011	$8(1) + 4(0) - 2(1) - 1(1) = 5$	0101 (5)
1100	$8(1) + 4(1) - 2(0) - 1(0) = 12$	Invalid
1101	$8(1) + 4(1) - 2(0) - 1(1) = 11$	Invalid
1110	$8(1) + 4(1) - 2(1) - 1(0) = 10$	Invalid
1111	$8(1) + 4(1) - 2(1) - 1(1) = 9$	1001 (9)

If the decimal equivalent is between 0 and 9, the code is valid.

Step 3: Karnaugh Maps (K-maps) for BCD Output

We derive Boolean expressions for BCD outputs W, X, Y, Z using K-maps.

For example:

$$W = A$$

$$X = A \cdot B + \overline{A} \cdot B \cdot \overline{C} + A \cdot \overline{B} \cdot C$$

$$Y = \text{(similar approach)}$$

$$Z = \text{(similar approach)}$$